Ancient Greek Science
600 BCE – 300 CE

Before

Civilizations before the Greeks developed agriculture, practiced medicine, discovered numbers, math and applied engineering and used calendars (lunar, solar).

Greeks used science not only as a practical tool but also to explain nature.

Greeks wanted strictly deductive proofs based on stated axioms

Greek Science

Assumed there is natural order and rational structure to nature.

- Nature is not arbitrary but follows certain laws.

Greeks used observation to support their theories.

- Unlike before (no theories, just observation)

Greek Philosophers

Empedocles proposed things made up of 4 elements; earth, air, water, fire. Almost implying states of matter.

Anaxagoras everything is made of countless seeds, infinite and imperceptible (Particles)

Democritus called this ATOM.

Aristarchus estimate the weights of Sun, Moon and Earth and proposed that in fact Earth went around Sun. *However, this was rejected as it conflicted with Aristotle's view.*

Thales

Thales discovered philosophy to answer existence, purpose of life, and truth. **Bridge for Science**

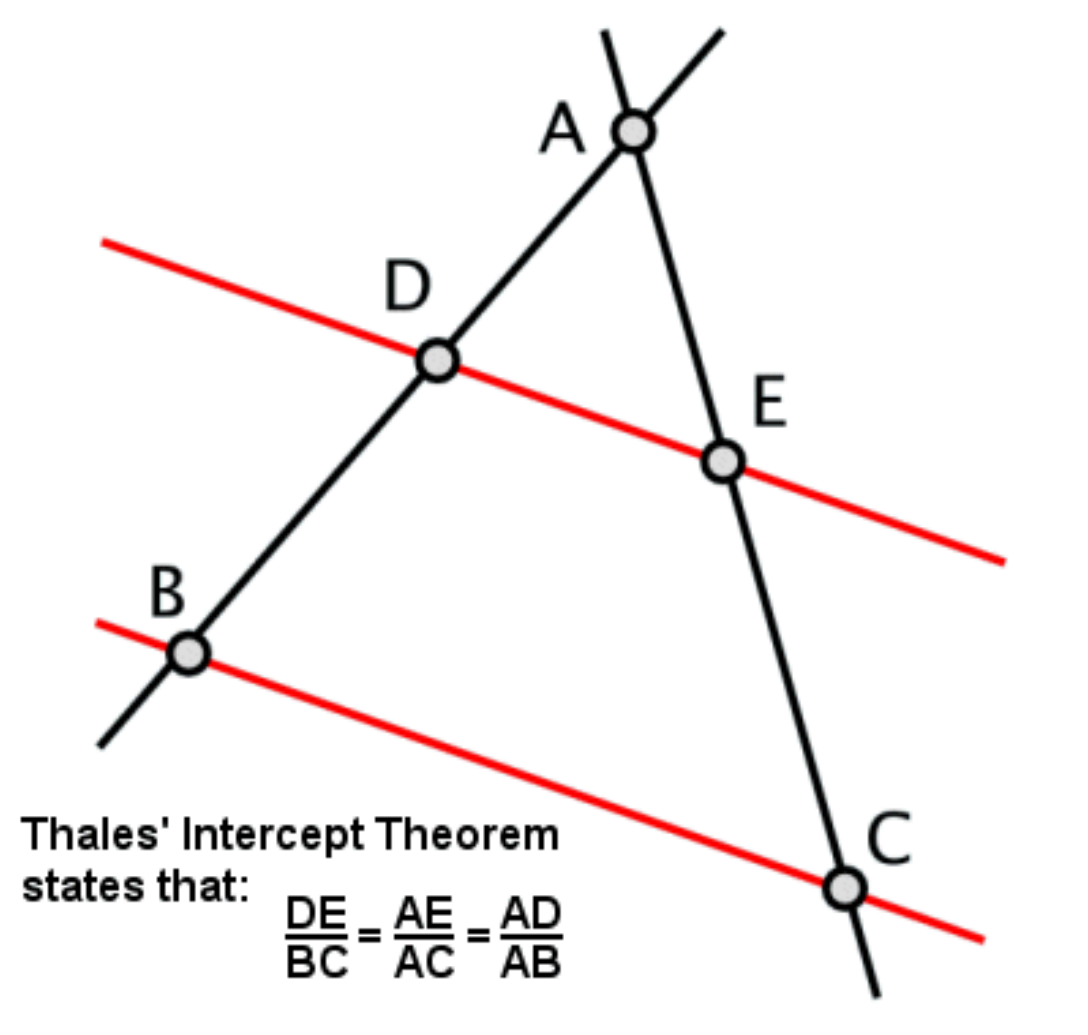
He claimed, living things came from *seeds* in moisture and they die in a *changing universe*.

Water was the ultimate substance.

He measured pyramids through shadow. (Geometry)

He thought Magnets have soul because they attract each other !!

Thales Intercept Theorem



Zeno's Paradox

All movements are illusion according to Parmenides. He built his philosophy based on **contradiction** (dispute Aristotle).

His student Zeno (in the absence of infinity) proposed paradoxes

1. Walking across a room but never reaching the wall

https://www.youtube.com/watch?v=MbNNFtuwA0k

2. Can Achilles catch Tortoise ?

This does not constitute a paradox as proven by the invention of **Calculus**

Pythagoreans seek power through mathematics

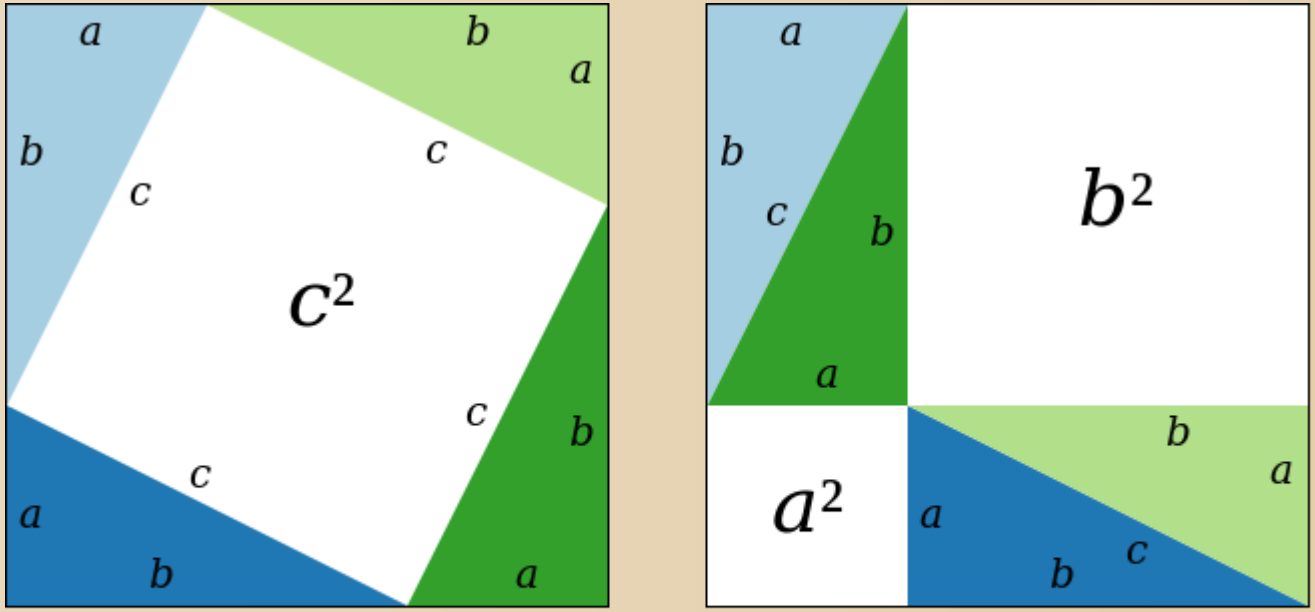
Pythagoreans thought even directly or by ratio anything can be represented.

They found new properties of **<u>numbers</u>**

- Perfect numbers: numbers that are the sum of their divisors
- Square numbers: *n* is equal to the sum of the first *n* odd numbers (e.g. $4^2 = 16 = 1 + 3 + 5 + 7$).
- Amicable numbers, Triangular numbers

Pythagoreans Theorem

Greeks were first to prove it.



$$c^2 = a^2 + b^2$$

Pythagorean theorem The sum of the areas of the two squares on the legs (a and b) equals the area of the square on the hypotenuse (c).

Irrational Numbers

Pythagoreans did not know the existence of irrational numbers.

Proof by contradiction, using Pythagorean theorem with a square of sides 1.

Prove $\sqrt{2}$ is irrational

~~Assume $\sqrt{2}$ is rational~~ $(\in \mathbb{Q})$ *

$\sqrt{2} = \frac{x}{y}$ $x, y \in \mathbb{Z}$, $y \neq 0$

x, y are in lowest terms

$\sqrt{2}\, y = x$

$2y^2 = x^2$ *

x even

$x = 2k$, $k \in \mathbb{Z}$

* $2y^2 = (2k)^2$

$2y^2 = 4k^2$

$y^2 = 2k^2$

y even

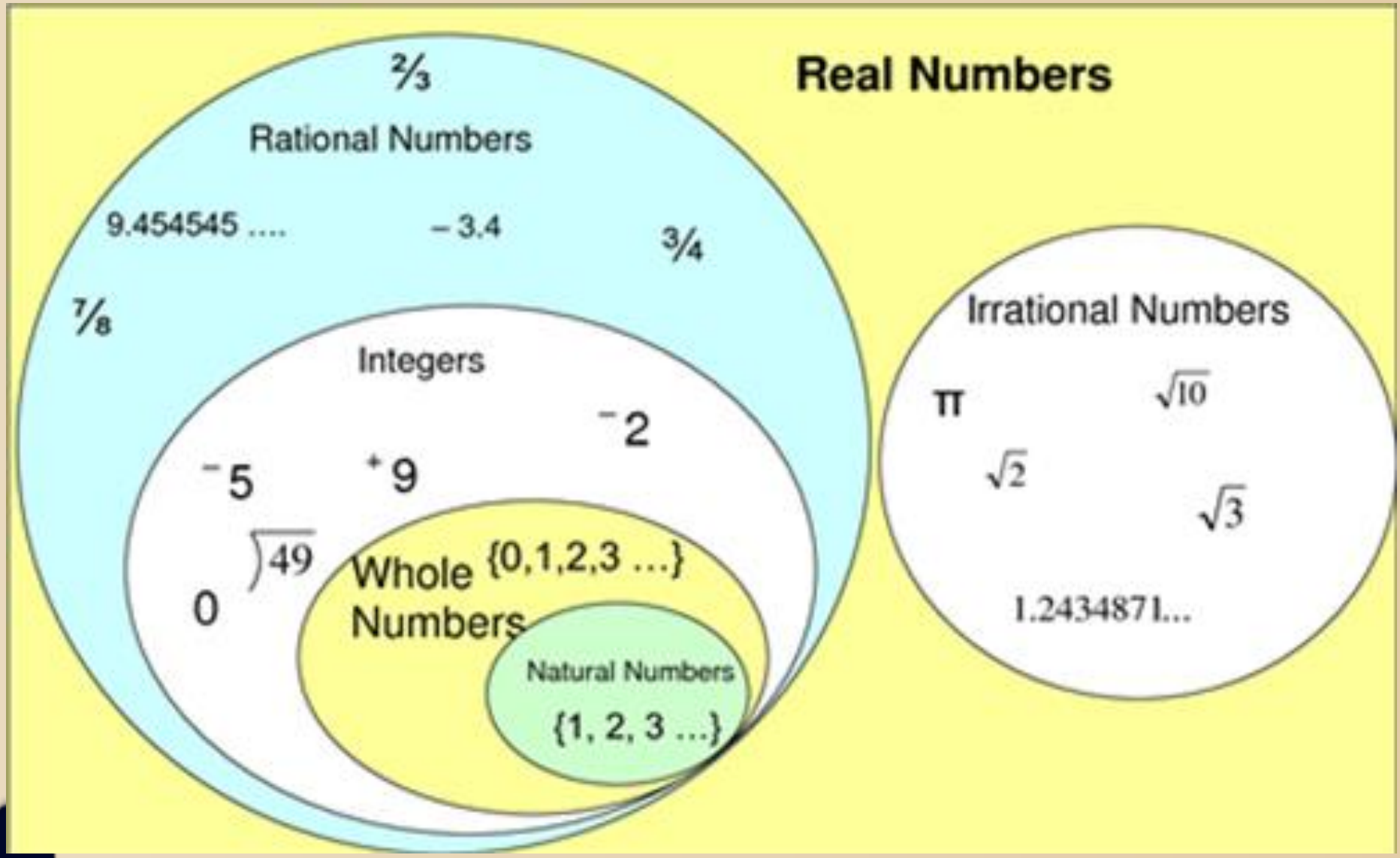
↯

Contradiction

$\sqrt{2}$

Irrational

Numbers


Real Numbers
2/3
Rational Numbers
9.454545 ....
– 3.4
3/4
7/8
Integers
⁻2
⁻5
⁺9
√49
0
Whole Numbers
{0,1,2,3 ...}
Natural Numbers
{1, 2, 3 ...}
Irrational Numbers
π
√10
√2
√3
1.2434871...

Euclid ~ 300 BC

He wrote 13 volume of Elements of Geometry book from known geometry of the time.

He gives definitions, axioms and mathematical proofs for postulates.

1st Postulate: Between 2 points a line can be drawn.

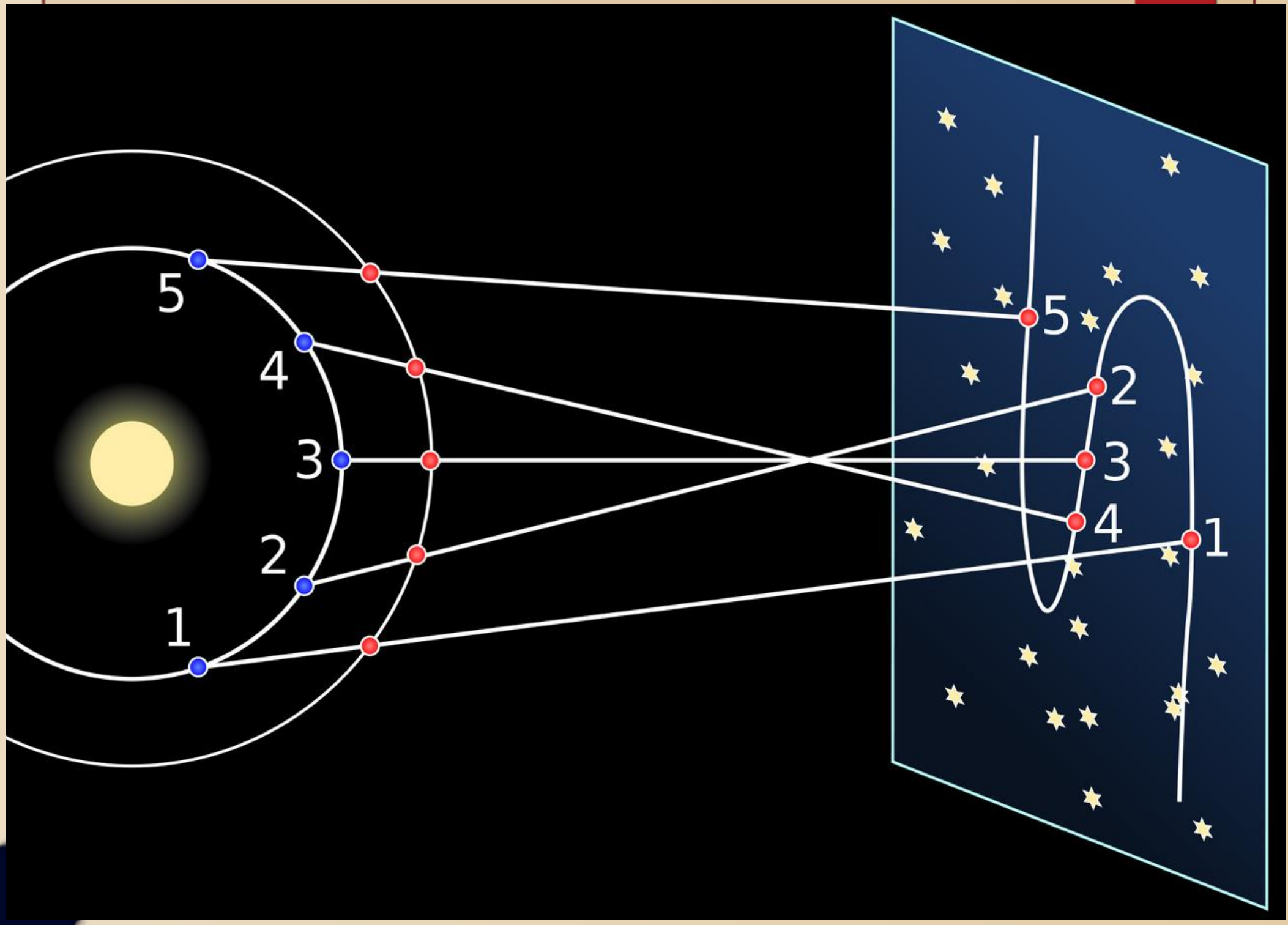
Circle/Sphere is Perfect for Greeks

They believed heavenly bodies are perfect. They must be spherical and moves in circles.

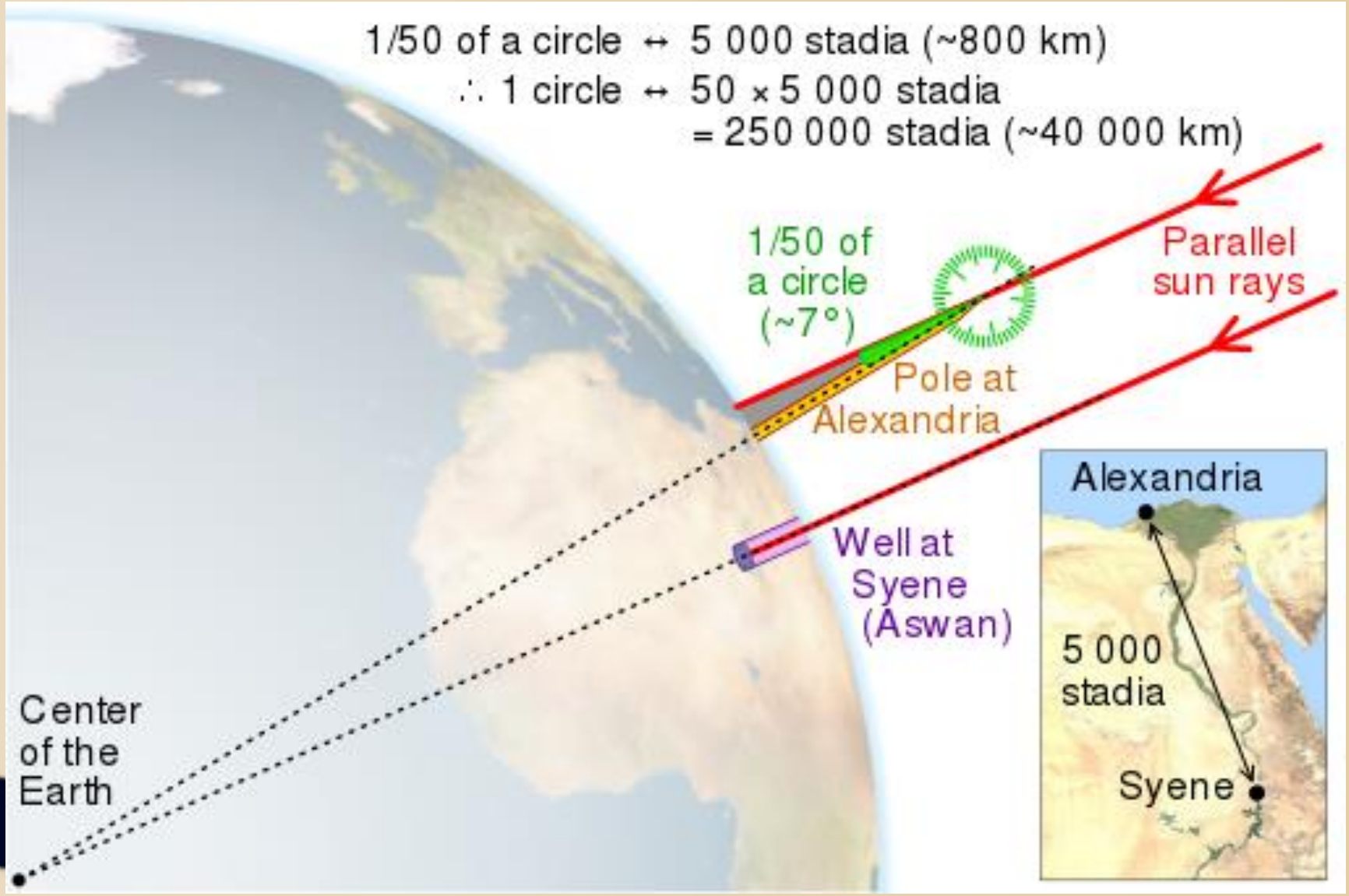
Ancient astronomers noted that planets would appear to move backwards against the background of stars periodically.

This Retro (backwards) grade (motion) did not conform well to the concept of an orderly universe ruled and created by a perfect being.

Copernicus solved the issue by realizing that the planets in fact did not move, but only appeared to do so from our point of view.


5
4
3
2
1
5
2
3
4
1

Radius of Earth from Circumference by Eratosthenes – 270 BC Measured at equator on 21st March

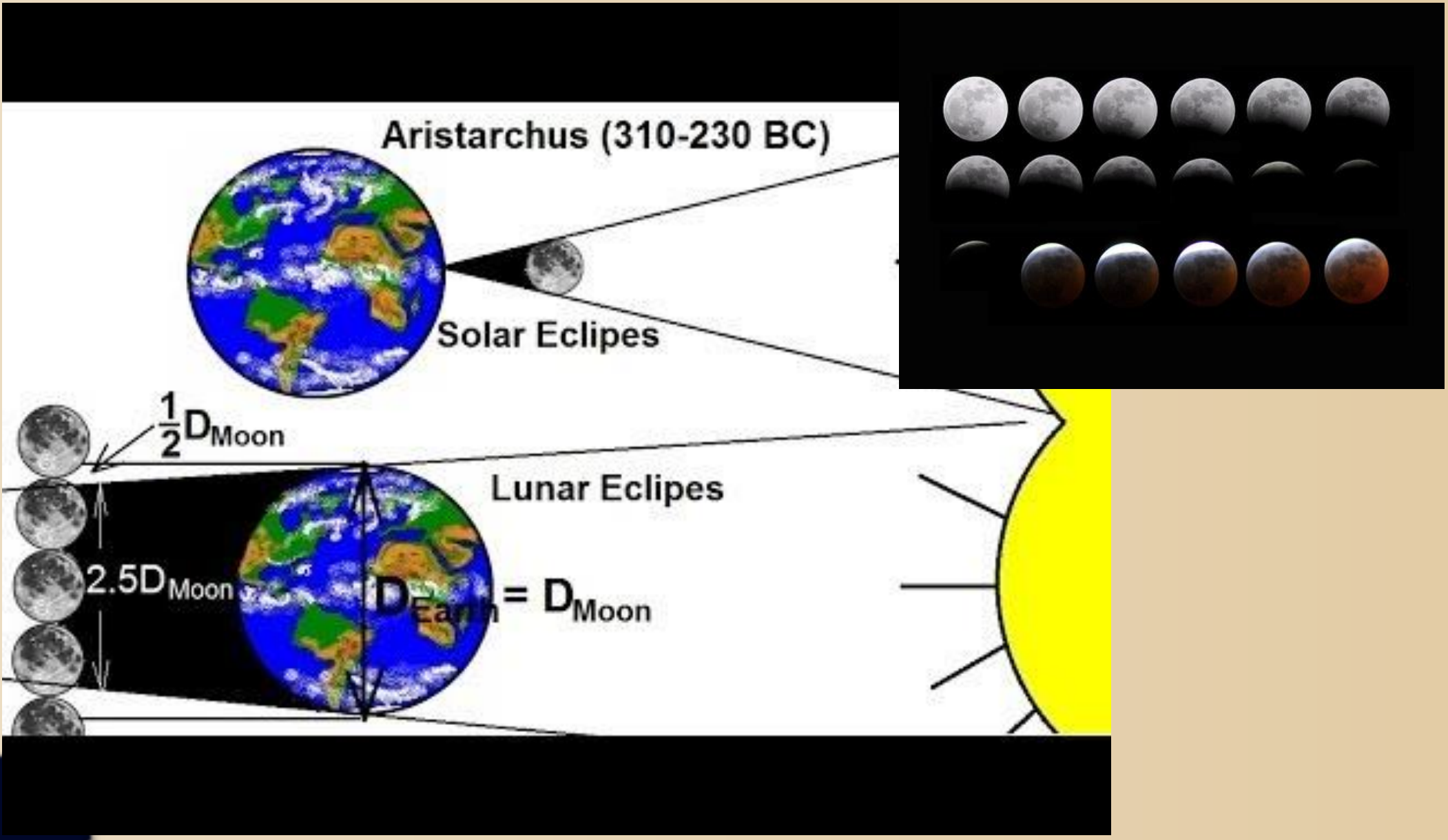


Aristarchus ~ 310-230 BC

Aristarchus used something he knew about the Earth (its size) and a bit of clever geometry to deduce the size of the Earth's shadow at the distance of the Moon.

Then by watching the Moon travel through the Earth's shadow (a lunar eclipse), he could determine the ratio between the size of the shadow at the distance to the Moon and the size of the Moon itself.

Size of Moon - Aristarchus



Distance from Moon

Aristarchus

DISTANCE MEASUREMENT

Moon artificial eclipse by 2 cents coin

Measurements : Coin Observateur distance = 2280 mm
Coin diameter = 19 mm

$$\frac{Moon\ diameter}{Coin\ diameter} = \frac{Moon\ Observateur\ distance}{Coin\ Observateur\ distance}$$

$$\frac{Moon\ Observateur\ distance}{Moon\ diameter} = \frac{2280}{19}$$

$$Moon\ Observateur\ distance = Moon\ diameter \times 120$$

Distance from Moon

Aristarchus – However he was off

FINAL CALCULATION

We measured :

$$Moon\ Observateur\ distance = Moon\ diameter \times 120$$

$$Moon\ diameter = \frac{Earth\ diameter}{3{,}79}$$

$$Earth\ diameter = 12551\ km$$

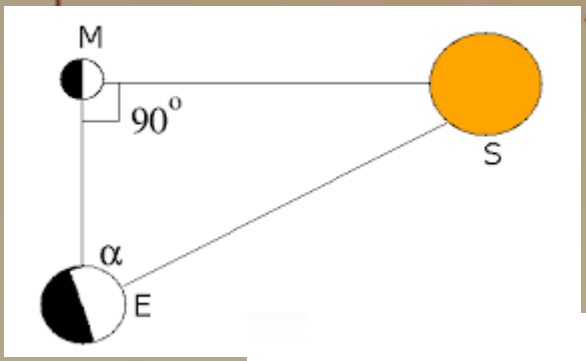
So :

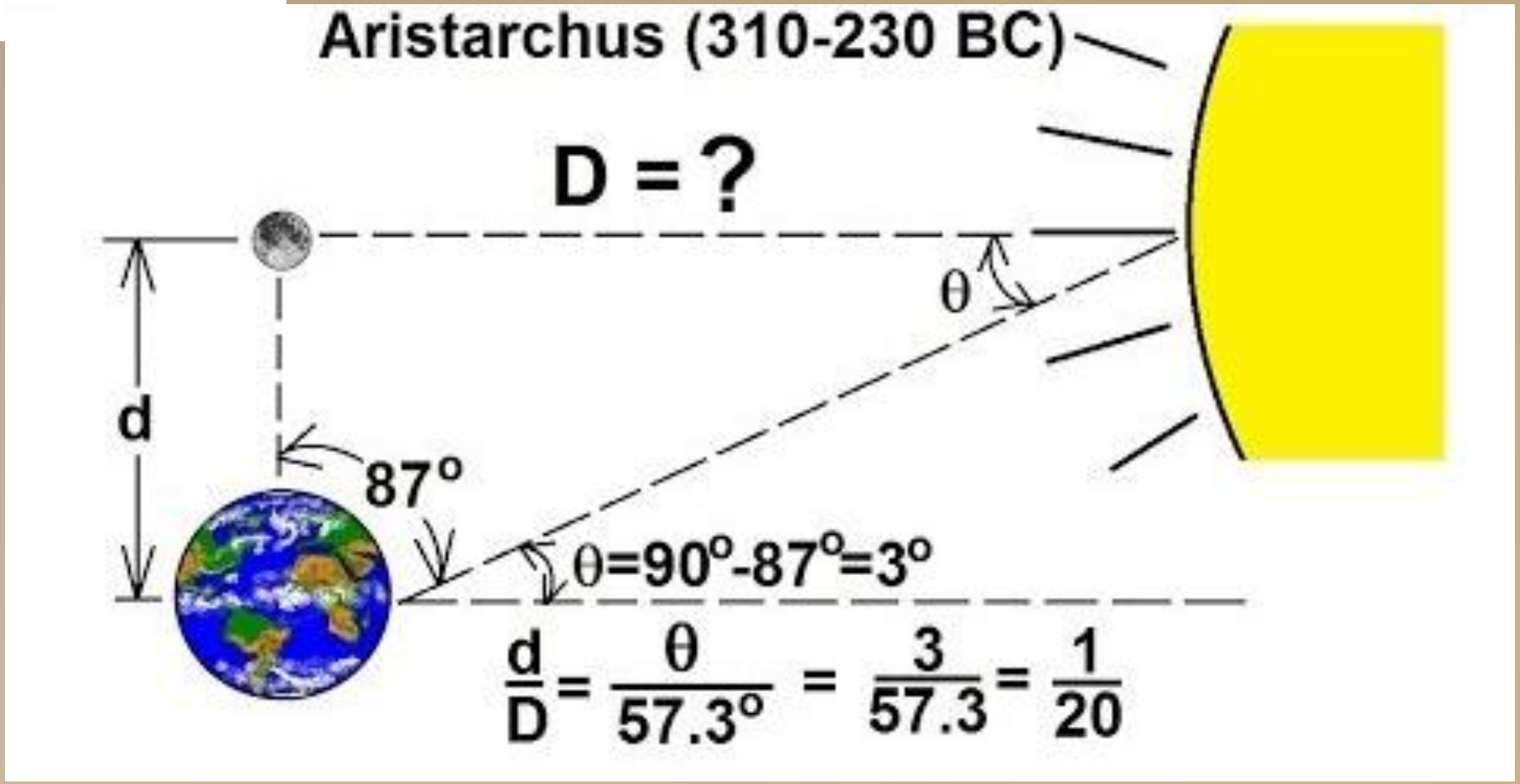
$$Moon\ Observateur\ distance = \frac{12551}{3{,}79} \times 120$$

$$\boxed{Moon\ Earth\ distance = 397393\ km}$$

Distance to Sun

Aristarchus





Motion of Aristotle

A body in motion seeks its own level unless a mover keeps acting upon it !! …

But what happens when a guy throws a ball in the air ...

Motion depends upon its weight and the density of the medium. !!

Archimedes - 250 BC

Great scientist even today. Preempted Newton in Mechanics, Physics. He invented physical modeling, using rigorous mathematical deductions from specified physical axioms.

He formulated the Law of the Lever, based on center-of-mass.

He anticipated methods of integral calculus.

Mechanical Method applied the Law of the Lever to determine the volume of a sphere

He estimated PI ~ 3.14.. to great accuracy.

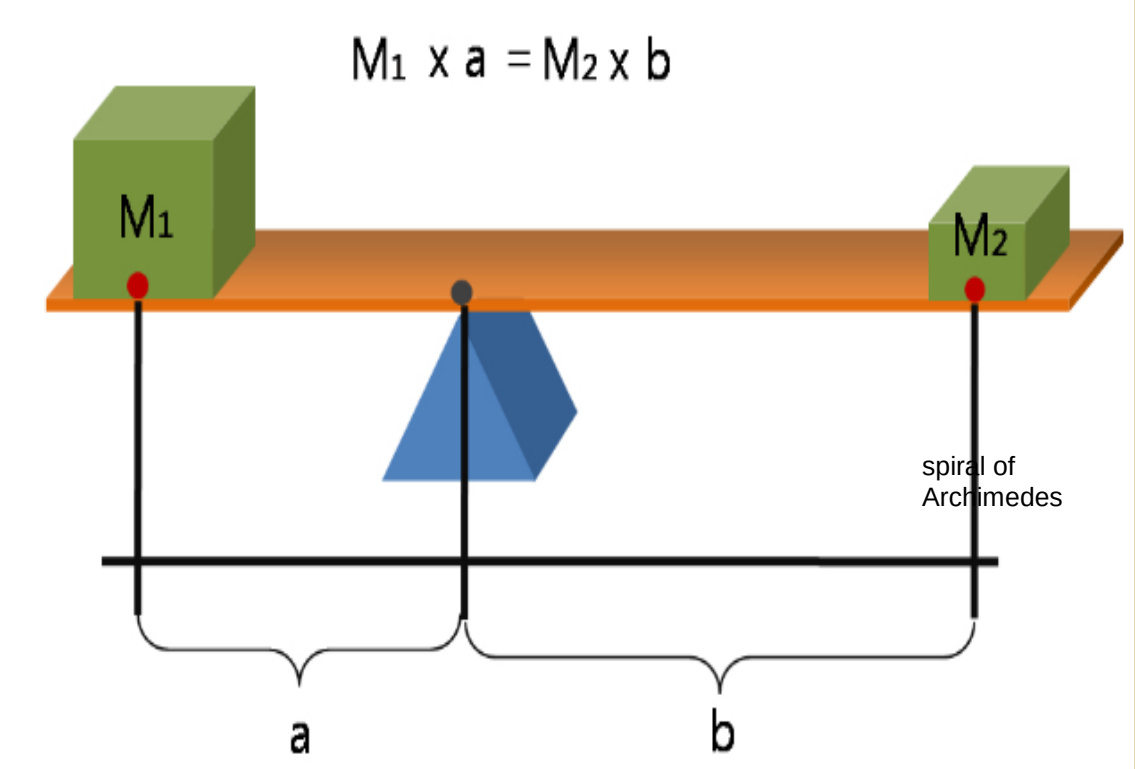
Spiral of Archimedes for pumping water

Defense Industry: The mirror + Catapults + Lifts for Ships

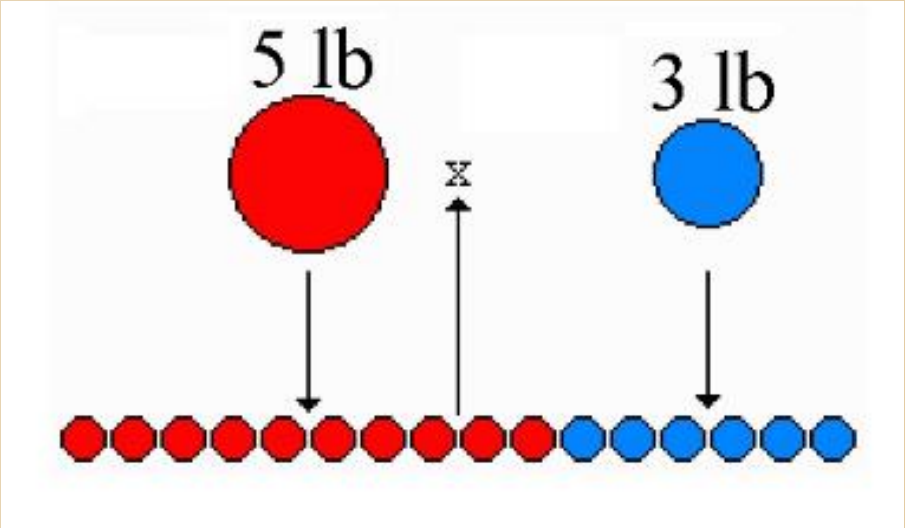
Law of Buoyancy

Law of Lever

Give me a lever long enough and a place to stand and I will move the world - Archimedes

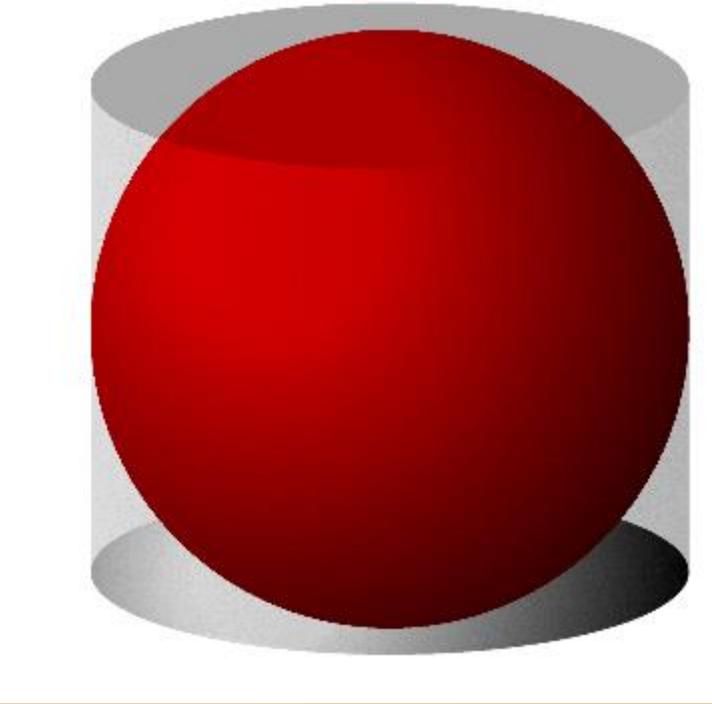


A distributed mass balances like the same mass concentrated at its center-of-mass.

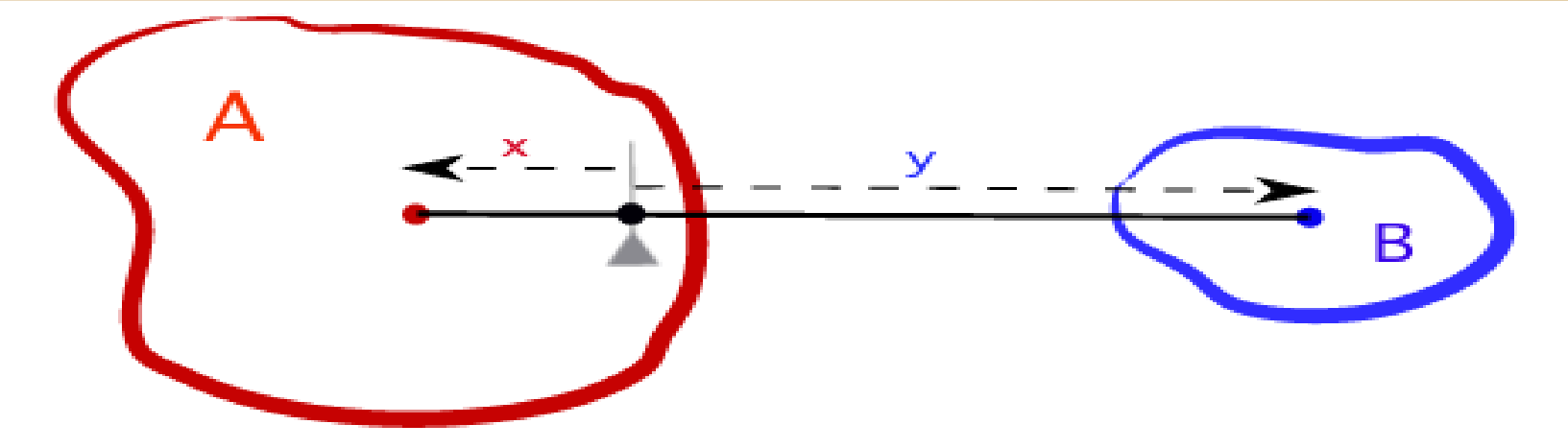


Archimedes' Proudest Achievement

The enclosed sphere has 2 / 3 the volume of the cylinder.



Axioms - Center-of-Mass for Areas



$$(Area\ A) \cdot x = (Area\ B) \cdot y$$

Deduction of Volume of Spheres

Using Law of Lever, Archimedes calculated the volume of Sphere.

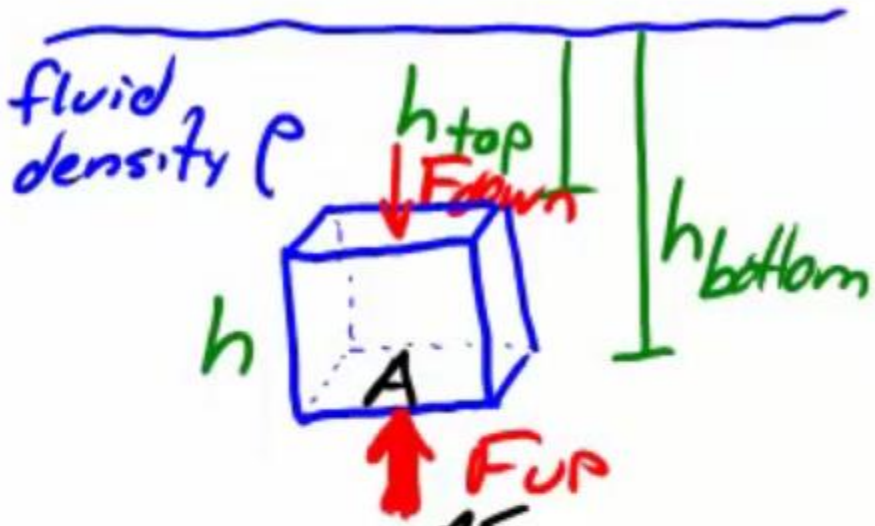
- Slides -

Law of Buoyancy

The volume of displaced fluid is equivalent to the volume of an object fully immersed in a fluid or to that fraction of the volume below the surface for an object partially submerged in a liquid.

The weight of the displaced portion of the fluid is equivalent to the magnitude of the buoyant force.

Buoyant Force



KEY IDEAS
you should already know

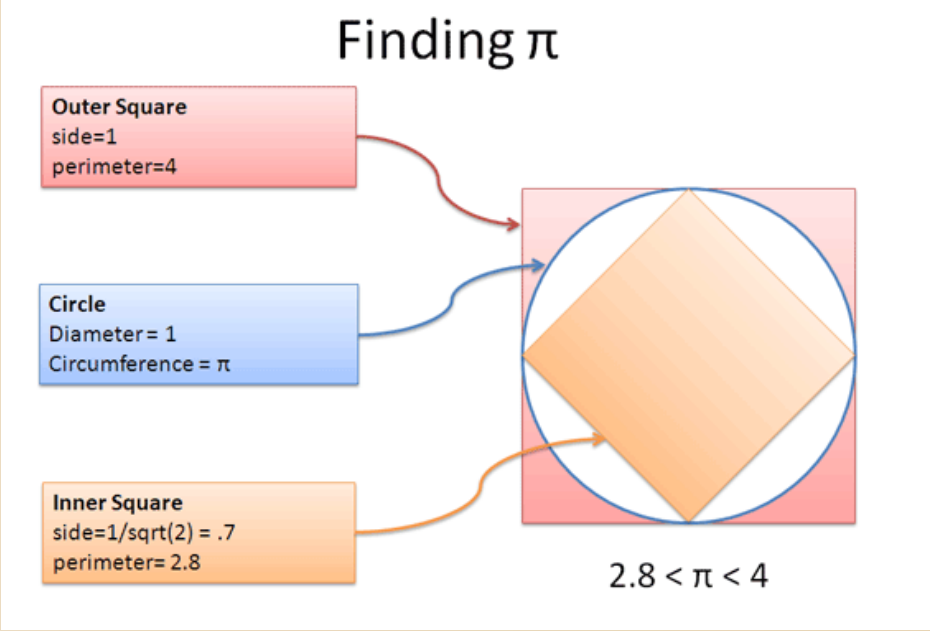
$P = \frac{F}{A} \therefore F = PA$

$P = \rho g h$

$\rho = \frac{m}{V} \therefore m = \rho V$

$$F_B = F_{up} - F_{down}$$
$$= P_{bottom} \cdot A - P_{top} \cdot A$$
$$= \rho g h_{bottom} A - \rho g h_{top} A$$
$$= \rho g A (h_{bottom} - h_{top})$$
$$= \rho g A h$$
$$F_B = \rho g V = m_{d.f.} g$$
$$F_B = F_{g,d.f.}$$

Finding PI



Archimedes didn't know the circumference of a circle. Started with what he did know: the perimeter of hexagons.

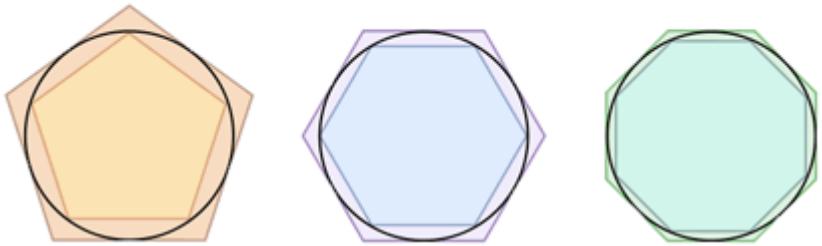
Let's draw it between two squares

Whatever the circumference is, it's somewhere between the perimeters of the squares:

- More than the inside,
- Less than the outside.

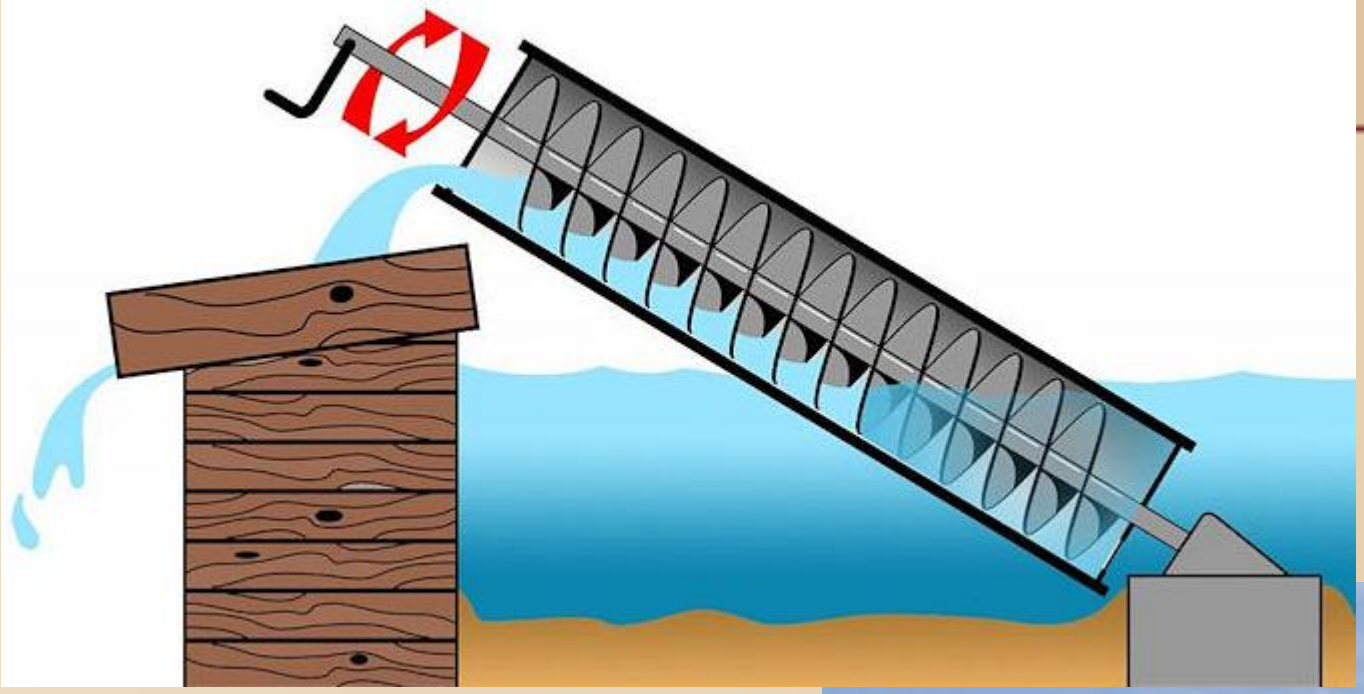
Foundations of Calculus

More Sides = Better "Circle"



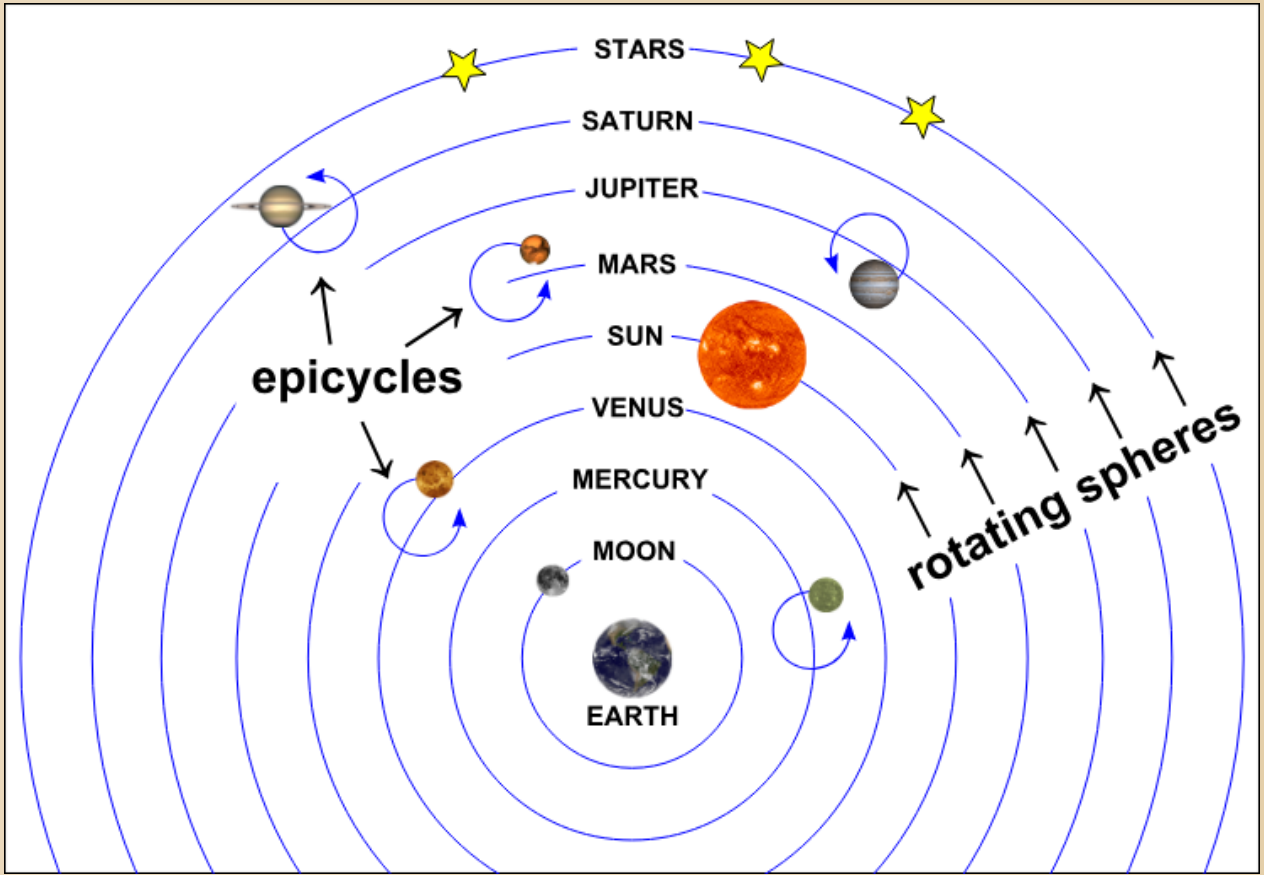
Sides	Inside	Outside	Pi guess	% Accuracy	Notes
4	2.828427125	4	3.414213562	91%	
8	3.061467459	3.313708499	3.187587979	98.535923322%	
16	3.121445152	3.182597878	3.152021515	99.668039026%	
32	3.136548491	3.151724907	3.144136699	99.919020520%	99.9% already!
64	3.140331157	3.144118385	3.142224771	99.979879075%	
128	3.141277251	3.14222363	3.14175044	99.994977489%	
256	3.141513801	3.141750369	3.141632085	99.998744854%	
512	3.14157294	3.141632081	3.141602511	99.999686244%	Five nines!
...					
524288	3.141592654	3.141592654	3.141592654	100.000000000%	Excel's limit

Archimedes Screw



Ptolemy's Geocentric

The astronomical predictions of Ptolemy's Geocentric model were used to prepare astrological and astronomical charts for over 1500 years.



Ptolemy's Geocentric

There was much resistance to the transition between these two theories. Christian theologians were reluctant to reject a theory that agreed with Bible passages.

Model assisted sailors for centuries in Mediterranean sea.

The Geocentric model held sway into the early modern age, but from the late 16th century onward was gradually superseded by the Heliocentric model of Copernicus, Galileo and Kepler.

Medical Developments

Hippocrates of Kos and Galen contributed to the medical field primarily through dissection.

Galen thought that imbalance in body fluids what caused disease.

His idea of circulation system was good but was not quite right.

- Push-Pull action of heart.
- The lungs provided cooling air. (Not right)

End of Greeks

Diophantus – Father of Algebra

- First to recognize fractions.
- He wrote a book Arithmetica that provided solutions to algebraic equations.

Roman empire ended progress. Lucretius (50 BC) wrote a poem where he translated most of Greek science titled “On the Nature of the Universe”

What the Ancients Knew - Greece

What the Ancients Knew The Greeks History Documentary

https://www.youtube.com/watch?v=XmgBOACkAzc